

Maso Mbali

Boksburg Campus

20241232

NET731

Practical Week 7

Task 1: Enable Azure Monitor

The screenshot displays the Microsoft Azure portal interface. At the top, the navigation bar shows 'Microsoft Azure' and the user's account information. The breadcrumb trail indicates the path: 'All services > Compute infrastructure | Virtual machines > CtuStudent-Vnet'. The main heading is 'CtuStudent-Vnet | Diagnose and solve problems', with 'Virtual machine' as a subtitle. A search bar is located at the top left of the main content area. On the left sidebar, the 'Diagnose and solve problems' option is selected, highlighted in blue. Below it, other options like 'Overview', 'Activity log', 'Access control (IAM)', 'Tags', 'Monitor', 'Resource visualizer', 'Connect', 'Networking', 'Settings', 'Disks', 'Extensions + applications', 'Operating system', 'Configuration', 'Advisor recommendations', 'Properties', 'Locks', 'Availability + scale', 'Security', 'Backup + disaster recovery', 'Operations', 'Monitoring', and 'Automation' are listed. The main content area is titled 'What problem are you trying to troubleshoot?' and includes a text input field for 'Briefly describe the issue' and a 'Go' button. Below this, there are two status boxes: 'No Azure health events detected. View service health' and 'No changes to this resource in the last 24 hours. View change details'. A section titled 'Common problems (10)' is also visible, with a description: 'Explore the most common problems for your resource. Select Troubleshoot to run an automated troubleshooter, follow do-it-yourself troubleshooting steps, or explore a wide range of troubleshooting tools.' At the bottom right, there is a 'Give feedback' link.

This is where I would have completed the practical if my subscription was active.

I would have enabled VM Insights and Diagnostic settings on my virtual machine to begin collecting essential telemetry like CPU, memory, and disk performance data. For my network resources, I would have configured diagnostic settings on the Virtual Network and Network Security Group to track traffic flow and security events, directing this data to a Log Analytics Workspace for centralized storage. I would have ensured that the correct log categories, such as network security group rule counters, were selected to provide a comprehensive view of the environment's health and security posture.

Task 2: Review Key Metrics

The screenshot shows the Microsoft Azure portal interface. The top navigation bar includes the Microsoft Azure logo, a search icon, and a user profile with the email 20241232@ctucareer.co... and the name MM. The left sidebar shows the 'Metrics' tab selected. The main content area is titled 'Select a scope' and features a 'Browse' tab. Below the tabs, there are dropdown menus for 'Resource types' (set to 'All resource types') and 'Locations' (set to 'All locations'). A search bar is present with the placeholder text 'Search to filter items...'. Below the search bar is a table with three columns: 'Scope', 'Resource type', and 'Location'. The table lists three items: 'Azure for Students' (Subscription), 'CTU-Student-RG' (Resource group), and 'NetworkWatcherRG' (Resource group). At the bottom of the dialog, there is a section for 'Selected scopes' which currently shows 'No scopes selected'. Below this, it says 'No results'. At the bottom right, there are buttons for 'Apply', 'Cancel', and 'Clear all selections'.

Scope	Resource type	Location
☐ Azure for Students	Subscription	-
☐ > CTU-Student-RG	Resource group	-
☐ > NetworkWatcherRG	Resource group	-

I would have used the Metrics Explorer to monitor CPU usage, selecting a specific time range to identify performance trends under different workloads. To track network throughput, I would have created a single chart showing both inbound and outbound traffic in bytes or bits per second to visualize the data flow patterns. Finally, I would have reviewed disk metrics by looking at read and write operations to understand how the system handles data-intensive tasks and to identify any potential storage bottlenecks.

Task 3: Document Your Monitoring Setup

I would have documented my configuration by listing every monitored resource and the specific data collection methods used, such as VM Insights for the server and diagnostic settings for the networking components. My summary would have explained that the data is stored in a Log Analytics Workspace, where it remains accessible based on the defined retention policy for historical analysis. I would have also reflected on the metrics observed, noting that consistent performance is expected and explaining how production anomalies, like unexpected traffic spikes, would be investigated using these tools.

Reflection and Alerts

I would have configured an alert rule to trigger a notification if CPU usage exceeded eighty percent for five minutes, allowing for proactive intervention before a performance issue impacted users. In my reflection, I would have explained that cloud monitoring is vital for visibility into virtualized infrastructure and that metrics are better for real-time performance tracking while logs are essential for deep-dive troubleshooting and security audits. I also would have noted that monitoring data is a key tool for cost optimization, as it reveals underutilized resources that could be resized to save money.